

Model Showdown

Income-Threshold Screening for Manual Benefit Verification

Comparing KNN, Logistic Regression and Random Forest



Decision context

The model predicts income > USD 50,000.

A positive result triggers manual benefit verification — never automatic rejection.

FALSE POSITIVE

Delay for an eligible applicant

FALSE NEGATIVE

Incorrect allocation of funds

PRIORITY

Precision chosen pre-training to
reduce wrongful flags

Dataset & limitations

48,842

original records

14

raw features

23.9%

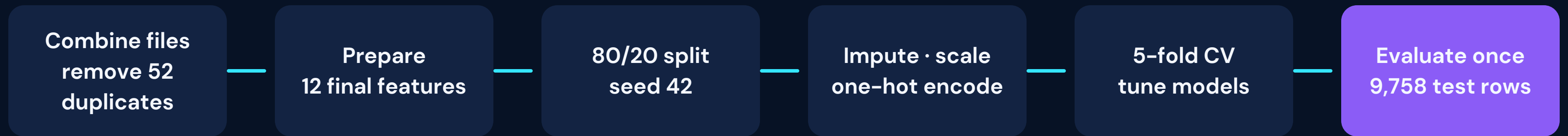
positive class (>50K)

CAUTION

Old · US-specific · imbalanced
Not representative of Dutch applicants

Method

A reproducible pipeline, with the test set opened once.



Model comparison

MODEL	PRECISION	RECALL	F1
Baseline	0.000	0.000	0.000
KNN	0.728	0.624	0.672
Logistic Regression	0.756	0.488	0.593
Random Forest	0.906	0.287	0.436

Precision is not enough

Precision versus recall

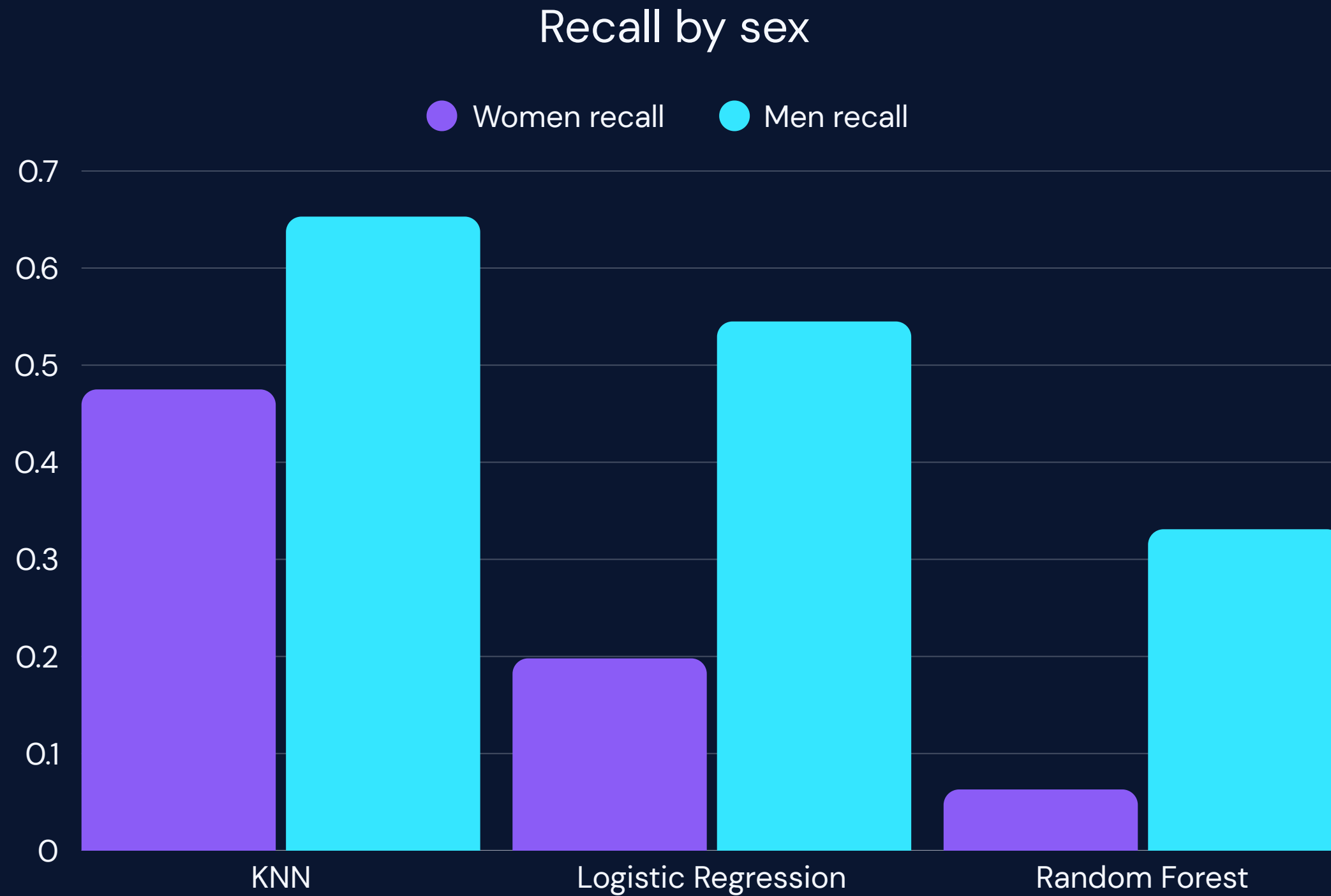
● Precision ● Recall



RF misses ~7/10 actual high-income cases.

KNN's F1 of 0.672 offers the strongest balance of usefulness and harm.

Fairness check



All models perform worse for women.

KNN has the smallest gender recall gap.
The disparity remains unacceptable for deployment.

Ethical risks

FAIR

Fairness

Historical inequality and proxy discrimination

AUTO

Autonomy

Notice, explanation and appeal rights

PRIV

Privacy

Collect only necessary data; protect it

ACCT

Accountability

The public organisation remains responsible

Protected attributes may support auditing — but should not automatically determine scrutiny.

Recommend KNN for the classroom experiment

Best tested balance of usefulness versus harm · least severe sex-recall gap

0.728

PRECISION

0.624

RECALL

0.672

F1

Least problematic does not mean fair or deployment-ready.

Safeguards before real use



- Current local data
- DPIA + legal review
- Remove protected attributes
- Intersectional fairness testing

- Threshold review
- Logging · security · drift
- Kill switch

The best score is not automatically the best decision.

KNN is the strongest tested trade-off

Fairness gap + outdated data prevent deployment

Human rights and evidence trump model convenience

Questions?